Unit 6 Pre-Test

Functions & Linear Relationships

Standards assessed: NC.8.F.1–F.5 · NC.8.EE.8

Name: _____ **Date:** _____ **Period:** _____

SCORE 3.0 Basic Understanding

NC.8.F.1 · NC.8.F.3 · NC.8.F.5 Identify functions and read slope and y-intercept from an equation.

• *I can decide whether a relation is a function and read slope and y-intercept from y = mx + b.*

Match each vocabulary term to its definition, then answer each question.

– function	**A.** Where a graph crosses the y-axis (the value of y when x = 0).
– slope	**B.** A relation in which each input has exactly one output.
– y-intercept	**C.** A point written (x, y) that gives a location on the graph.
– ordered pair	**D.** The steepness of a line, found as rise over run.
– linear function	**E.** A function whose graph is a straight line.
– input	**F.** The x-value that is put into a function.

Slope-intercept form — reference

$$y = mx + b$$

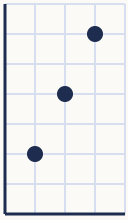
m = *slope (rise over run)* · **b** = *y-intercept (where x = 0)*

1. *Function?*

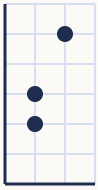
Is this a function? (1, 2), (2, 4), (3, 6)

2. *Function?*

Is this a function? (1, 2), (1, 3), (2, 5)



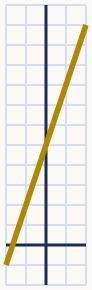
Answer: _____



Answer: _____

3. *Slope*

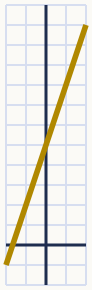
In y = 3x + 5, what is the slope?



m = _____

4. *y-intercept*

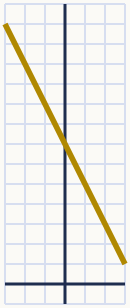
In y = 3x + 5, what is the y-intercept?



b = _____

5. *Slope*

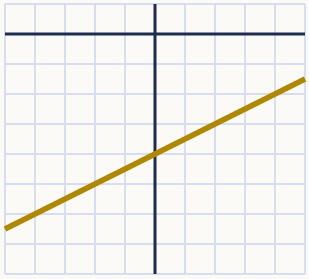
In y = −2x + 7, what is the slope?



m = _____

6. *y-intercept*

In y = ½x − 4, what is the y-intercept?

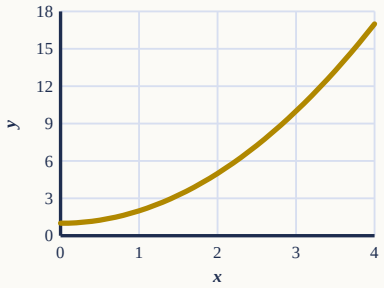


b = _____

7. *Linear?*

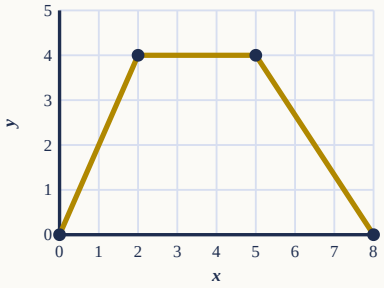
Is $y = x^2 + 1$ linear or nonlinear?

8. *Read a graph*



Answer: _____

Look at the graph. From x = 2 to x = 5, is the graph increasing, decreasing or constant?



Answer: _____

SCORE 4.0 Target Content

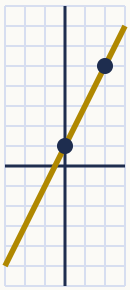
NC.8.F.3 · NC.8.F.4 · NC.8.F.5 Find slope and y-intercept from two points and from a table.

• *I can find the slope and y-intercept from two points or a table of values.*

Find each slope or y-intercept.

9. *Slope*

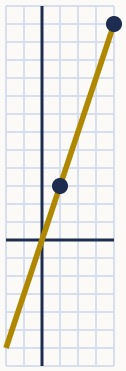
Find the slope through (0, 1) and (2, 5).



m = _____

10. *Slope*

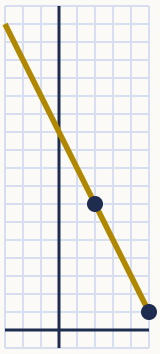
Find the slope through (1, 3) and (4, 12).



m = _____

11. *Slope*

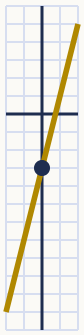
Find the slope through (2, 7) and (5, 1).



m = _____

12. *y-intercept*

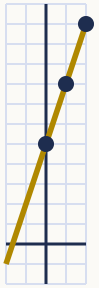
A line has slope 4 and passes through (0, −3). Find the y-intercept.



b = _____

13. *From a table*

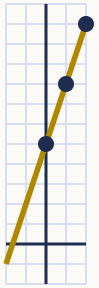
x: 0, 1, 2 and y: 5, 8, 11. Find the slope.



m = _____

14. *From a table*

x: 0, 1, 2 and y: 5, 8, 11. Find the y-intercept.



b = _____

15. *From a table*

Is this table linear or nonlinear?

x	0	1	2	3
y	1	2	5	10

Answer: _____

16. *From a table*

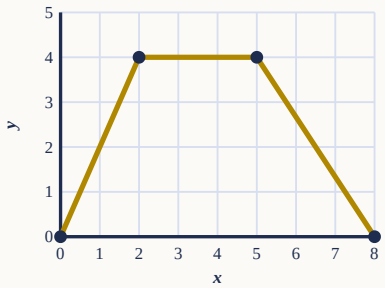
Is this table linear or nonlinear?

x	0	2	4	6
y	3	7	11	15

Answer: _____

17. *Read a graph*

On which part is the graph decreasing?



Answer: _____

SCORE 5.0 Advanced Content

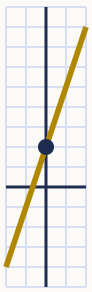
NC.8.F.2 · NC.8.F.3 · NC.8.F.5 · NC.8.EE.8 Write equations, compare functions, interpret slope, and solve a system by graphing.

• *I can write a linear equation, compare two functions, interpret slope as a rate of change, and read the solution of a system from its graph.*

Answer each question.

18. *Write equation*

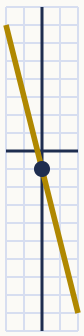
A line has slope 3 and passes through (0, 2). Write y = mx + b.



y = _____

19. *Write equation*

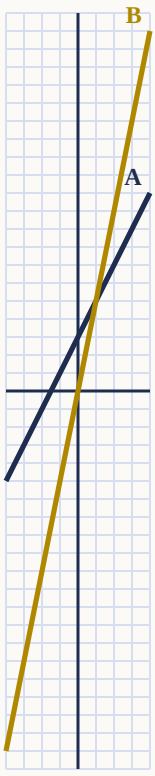
A line has slope −4 and passes through (0, −1). Write y = mx + b.



y = _____

20. *Compare*

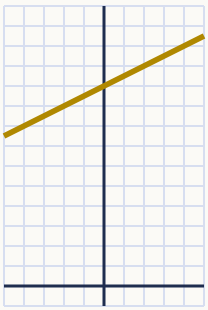
Function A: y = 2x + 3. Function B: slope 5. Which has the greater slope? (A or B)



Answer: _____

21. *Rate*

In y = 0.5x + 10, how much does y increase when x goes up by 1?



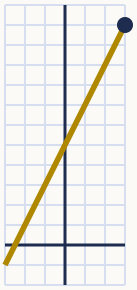
y increases by _____

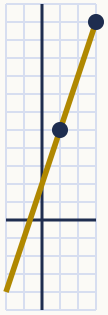
22. *Slope*

Find the slope through (1, 5) and (3, 11).

23. *y-intercept*

A line has slope 2 and passes through (3, 11). Find the y-intercept.



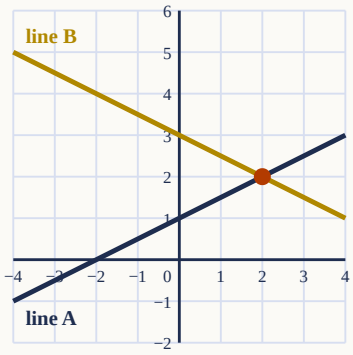


m = _____

b = _____

24. *System*

Lines A and B are graphed. Write the solution of the system as an ordered pair.

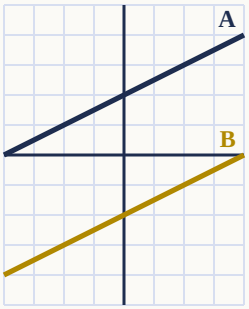


1 square = 1 unit

solution = _____

25. *System*

Lines A and B are graphed. How many solutions does this system have?



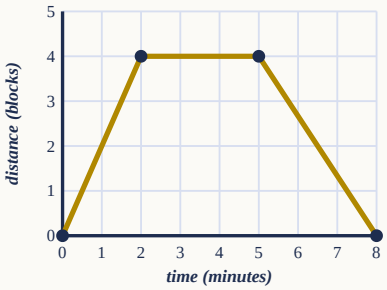
Answer: _____

26. *Graph story*

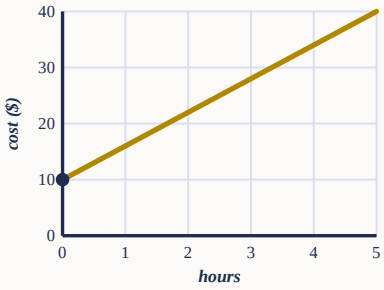
Mia walks to the store, shops, and walks home. The graph shows her distance from home. What does the flat part mean?

27. *Slope in a story*

A bike rental costs y = 6x + 10 dollars for x hours. What do the 6 and the 10 mean?



Answer: _____



Answer: _____